Cartier divisor (over any scheme X)

invertible sheaves + rational functions.

Cartier divisor $\longrightarrow$ invertible sheaves of X.

$$D \longmapsto \mathcal{L}(D).$$

Cartier divisor / linear equivalence $\cong$ invertible sheaves up to isomorphism.

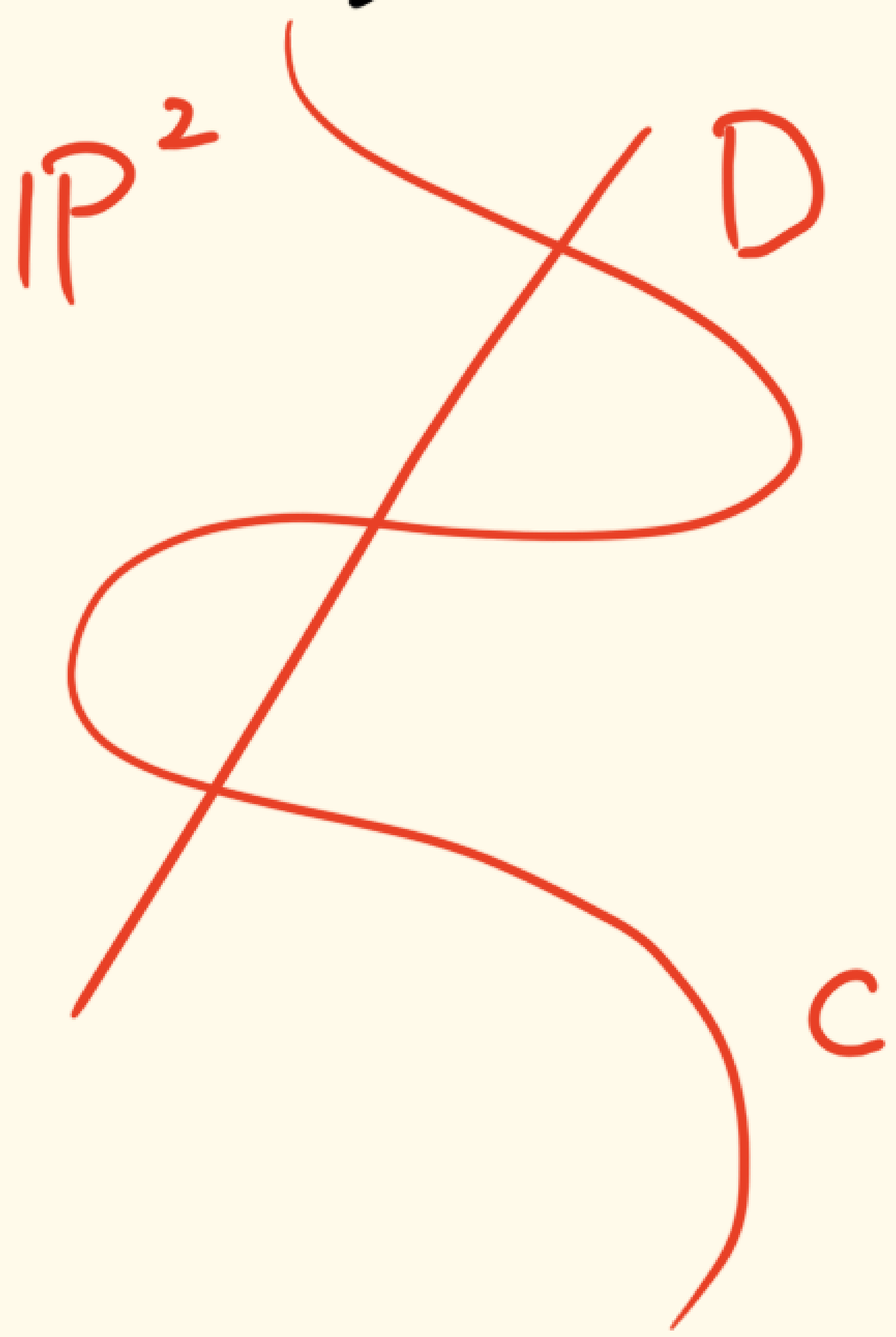
intersection product.

4.2. **Projective varieties and Lorentzian polynomials.** Let Y be a d-dimensional irreducible projective variety over an algebraically closed field $\mathbb{F}$. If $\mathrm{D}_1, \ldots, \mathrm{D}_d$ are Cartier divisors on Y, the *intersection product* $(\mathrm{D}_1 \cdot \ldots \cdot \mathrm{D}_d)$ is an integer defined by the following properties:

- the product $(\mathrm{D}_1 \cdot \ldots \cdot \mathrm{D}_d)$ is symmetric and multilinear as a function of its arguments,
- the product $(\mathrm{D}_1 \cdot \ldots \cdot \mathrm{D}_d)$ depends only on the linear equivalence classes of the D_i, and
- if $\mathrm{D}_1, \ldots, \mathrm{D}_n$ are effective divisors meeting transversely at smooth points of Y, then

$$(\mathrm{D}_1 \cdot \ldots \cdot \mathrm{D}_d) = \#\mathrm{D}_1 \cap \ldots \cap \mathrm{D}_d.$$

nef divisor: A $\mathbb{Q}$-divisor D is said to be nef if $(D\cdot C)\geq 0$ for any irreducible curve $C\subseteq X$. (X projective scheme).



Bezóut lemma

$D\cdot C = \deg D \cdot \deg C \geq 0.$

very ample relative to Y. X is a schemme. (2.5)

a sheaf $\mathcal{L}$ on X is very ample relative to Y if there

X is closed subscheme of an open subscheme of $\mathbb{P}^n_Y$

is an immersion $i: X \longrightarrow \mathbb{P}^n_Y$ for some n s.t.

$\mathcal{L} \cong i^*(\mathcal{O}(1))$, where $\mathbb{P}^n_Y := \mathbb{P}^n_A \times_{\operatorname{Spec} A} Y$.

fibre product.

$\exists$ a morphism $X \longrightarrow Y$. (X over Y).

$$\begin{tikzcd} X \arrow[r] \arrow[rr, bend left, "i"] & \mathcal{U} \arrow[r] & \mathbb{P}^n_Y \end{tikzcd}$$

closed immersion

open immersion.

morphism φ: scheme $X \longrightarrow \mathbb{P}^r_A$?

φ is uniquely determined by an invertible sheaf $\mathcal{L}$ and a set of its global sections. (2.7)

ample invertible sheaf. invertible sheaf $\mathcal{L}$ on a Noetherian scheme is ample if every coherent sheaf $\mathcal{F}$ on X, $\exists\, n_0 > 0$ (depending on $\mathcal{F}$) s.t. $\forall n \geq n_0$, $\mathcal{F} \otimes \mathcal{L}^n$ is generated by its global sections.

ex. X is affine, every coherent sheaf is generated by its global sections $\Rightarrow$ any invertible sheaf is ample. $n_0 = 0$.

ex. Serre's Thm: every very ample sheaf $\mathcal{L}$ on a projective scheme X over a Noetherian ring A is ample.

$\exists ! \, X \to \operatorname{Spec} A$.

Lemma. $\mathcal{L}$ is an ample invertible sheaf on a projective scheme iff $\mathcal{L}^{\otimes n}$ is very ample $\exists\, n$.

every nef divisor is a limit of ample divisors. ($\mathbb{Q}$-divisors).

topology ?

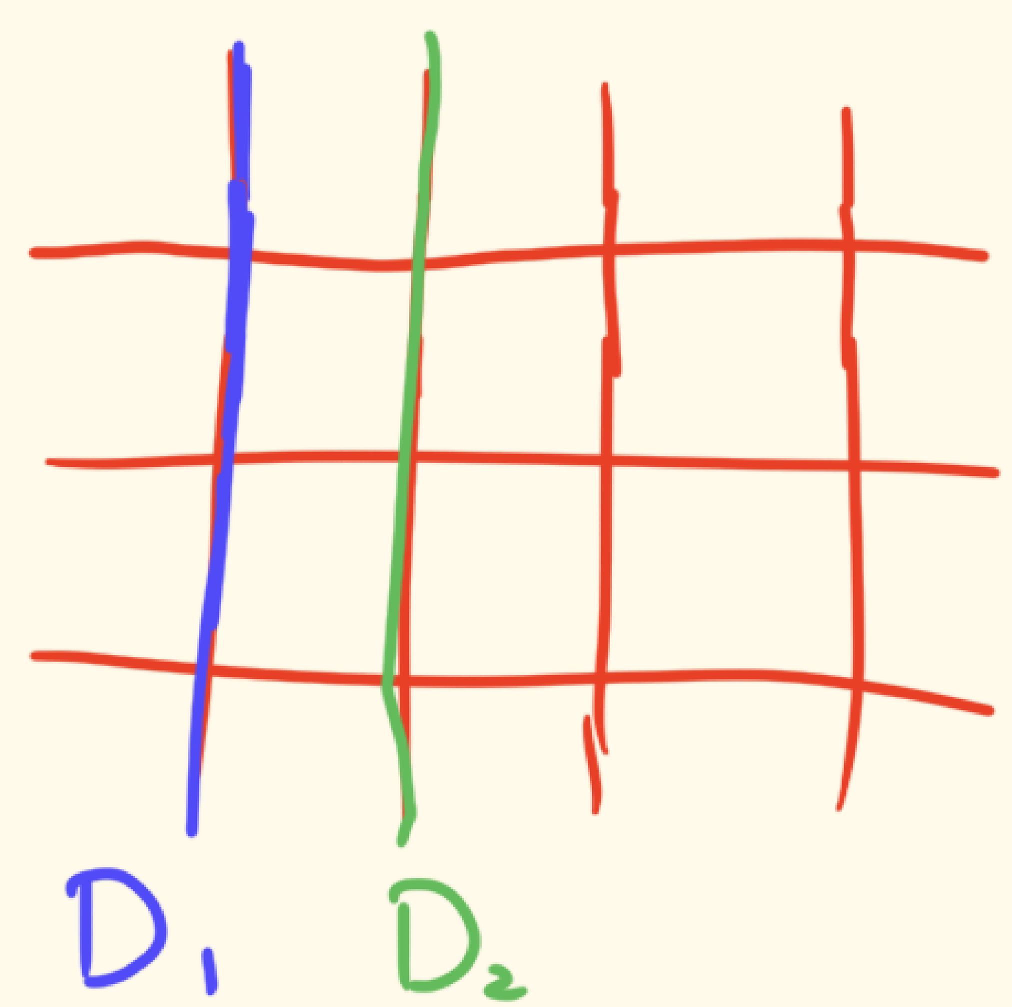
$\mathbb{Q}$-divisors $/ \sim$ =: $N^1(X)_{\mathbb{Q}}$ standard notation.

Euclidean top.

$\sim$: $\forall$ irreducible curve $C \subseteq Y$, we have $(D_1 \cdot C) = (D_2 \cdot C)$, $D_1 \sim D_2$ numerical equivalence.

Néron-Severi space over $\mathbb{Q}$. $\rightsquigarrow$ Euclidean top.

$X = \mathbb{P}^1 \times \mathbb{P}^1$

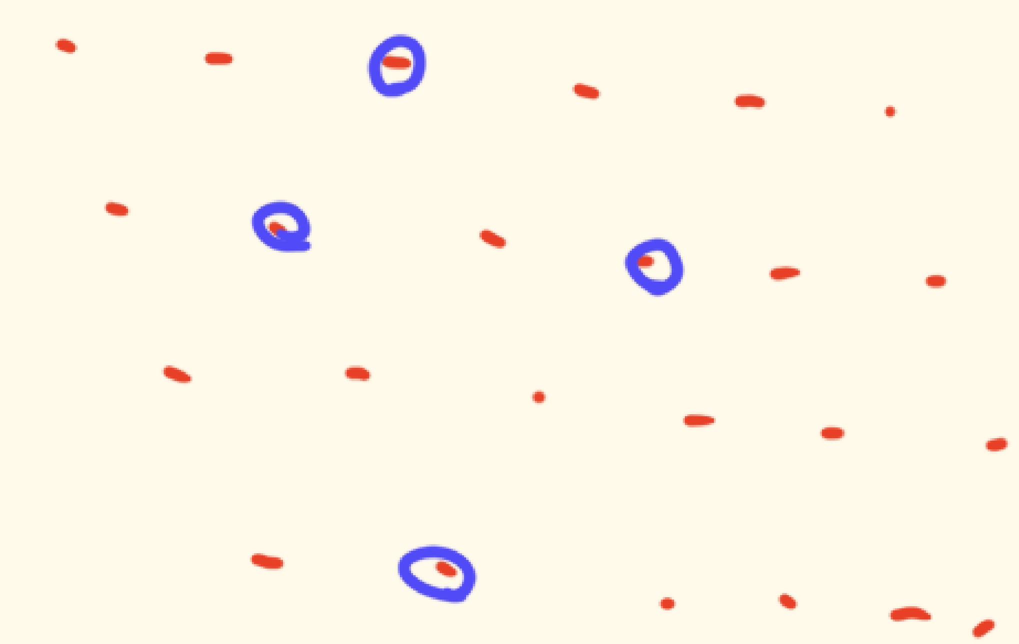


$D_1 \sim D_2$.

ample cone $\mathrm{Amp}(X)$ invertible sheaves.

$$\mathrm{Amp}(X) := \Big\{ \sum_{\text{finite}} q_i [\mathcal{L}_i] \in N^1(X)_{\mathbb{Q}} : q_i \in \mathbb{Q}.$$

$\mathcal{L}_i$ ample invertible sheaves $\Big\}$.



is open subset.

nef cone $\mathrm{Nef}(X)$ is closed subset.

Kleiman's Thm (2017) $\mathrm{Nef}(X) = \overline{\mathrm{Amp}(X)}$.

L_n^d is a closed subset of H_n^d. $L_n^d = \overline{(\mathring{L}_n^d)}$.

It suffices to show H_i is ample.

$$\sum_{i=1}^{n} \underset{\substack{\uparrow \\ \text{ample}}}{w_i H_i} . \quad \underset{\substack{\uparrow \\ \text{very ample}}}{H_i'} = M H_i \qquad \sum_{i=1}^{n} w_i H_i = \sum_{i=1}^{n} \left(\frac{w_i}{M}\right) \underset{\substack{\uparrow \\ \text{very ample.}}}{H_i'}$$

M is large enough.

It suffices to show H_i is very ample

every coefficient of vol_H is positive.

It is enough to show $\partial^\alpha \mathrm{vol}_H$ is Lorentzian $\forall \alpha \in \Delta_n^{d-2}$.

$$\frac{2!}{d!}\partial^\alpha \mathrm{vol}_H(\omega) = \Big(\sum_{i=1}^n \omega_i H_i \cdot \sum_{i=1}^n \omega_i H_i \cdot \underbrace{H_1 \cdots H_1}_{\alpha_1} \cdots \underbrace{H_n \cdots H_n}_{\alpha_n}\Big).$$

$$\mathrm{vol}_H(\omega) = (\omega_1 H_1 + \cdots + \omega_n H_n)^d = \sum_{\alpha \in \Delta_n^d} \frac{d!}{\alpha!} V_\alpha(H)\,\omega^\alpha.$$

$$V_\alpha(H) = (\underbrace{H_1 \cdots H_1}_{\alpha_1} \cdots \underbrace{H_n \cdots H_n}_{\alpha_n}).$$

$\exists$ irreducible surface $S \subseteq Y$ s.t. $\Downarrow$ Bertini's theorem

$$\frac{2!}{d!}\partial^\alpha \mathrm{vol}_H(\omega) = \Big(\sum_{i=1}^n \omega_i H_i \cdot \sum_{i=1}^n \omega_i H_i \cdot S\Big).$$

Hodge index thm. quadratic form has exactly one positive eigenvalue.